Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 35

Combinations

Covers Chartrand & Zhang, Section 8.4

Status of this set

Ungraded practice. Solutions released with it. Show the expression.

Part I: How to Practice This Material

The two formulas, and the one question

$$P(n,r) = \frac{n!}{(n-r)!} \qquad C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{P(n,r)}{r!}$$

Ask: does reordering change the outcome? Yes ⇒ permutation. No ⇒ combination.

Two shortcuts

$\binom{n}{r} = \binom{n}{n-r}$, so compute whichever is easier. And when a condition is awkward — "at least one" — count the complement instead.

Three mistakes to expect

1. **Using P where C belongs**, over-counting by $r!$.
2. **Adding when the stages are independent**, or multiplying when the cases are exclusive.
3. **Computing "at least one" by adding overlapping cases.** Use the complement.

Part II: Practice Problems

Section A — Which is it?

1. For each, state whether it is a permutation or a combination, and compute it.

(a) Choosing 3 pizza toppings from 8.

(b) Choosing a batting order for 9 of 12 players.

(c) Choosing 5 books from 11 to take on a trip.

Section B — Combining principles

2. A committee of 4 is chosen from 6 seniors and 5 juniors.

(a) How many committees are possible?

(b) How many have exactly 2 seniors?

(c) How many have at least 1 junior?

3. From a standard 52-card deck, how many 5-card hands

(a) consist entirely of hearts?

(b) contain exactly 3 hearts?

(c) contain at least one ace?

Section C — Recognising a hidden choice

4. Count each.

(a) Bit strings of length 12 with exactly five 1s.

(b) Paths on a grid from $(0, 0)$ to $(5, 3)$ moving only right or up.

(c) Handshakes if everyone in a group of 8 shakes hands once with everyone else.

Section D — Writing

5. In four to six complete sentences, explain why $C(n, r) = P(n, r)/r!$, and why $\binom{n}{r} = \binom{n}{n-r}$ can be seen without computing either side.

Section E — Challenge (optional)

6. A multiple-choice test has 10 questions, each with 4 options. How many ways are there to answer the test so that exactly 7 answers are correct?